

Order Tracking of Mobius Propeller Data

The Matlab code for this problem is listed in Appendix 1.

Sound Pressure Level Versus Microphone Placement Angle

Figure 1 shows the sound pressure generated by the Mobius propeller versus the angle of microphone placement.

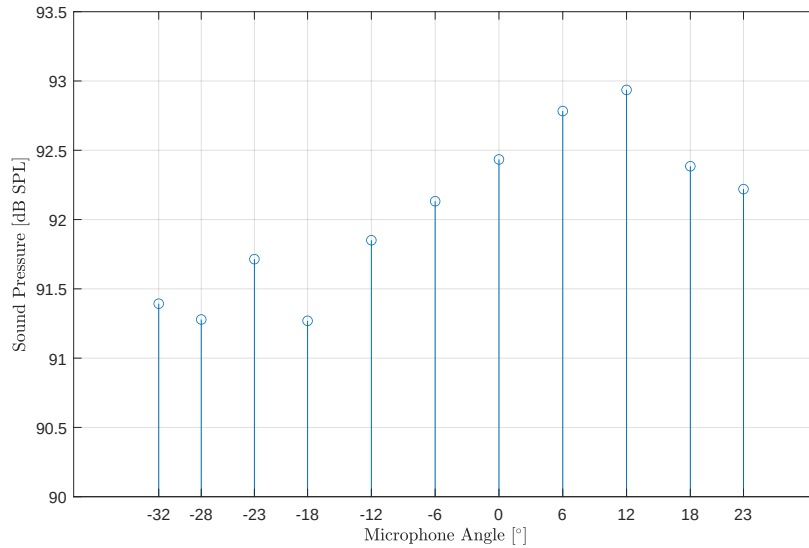


Figure 1: Sound pressure level versus microphone placement angle.

The highest sound pressure levels occur around the 0° plane, when the propeller is in the same plane as the microphone. The highest sound pressure, approximately 93 dB SPL, occurs at 12° , is radiated above the plane of the propeller and the drone itself.

Trigger Signal and Representative Pressure Recording

Figure 2 shows the angular position trigger signal and the pressure recording for a measurement angle of 0° .

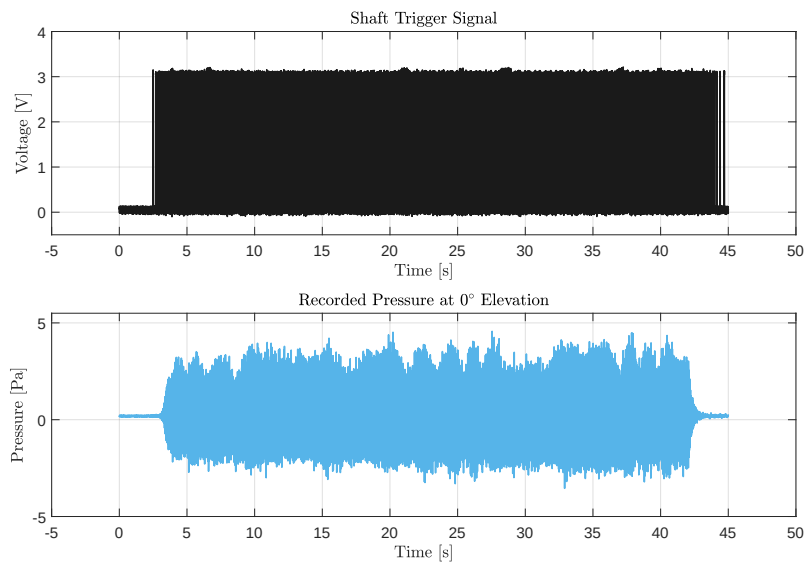


Figure 2: Trigger signal and a pressure recording at 0° .

Estimate of Shaft Rotation Speed

Figure 3 shows the estimate rotation speed of the shaft per revolution of the shaft.

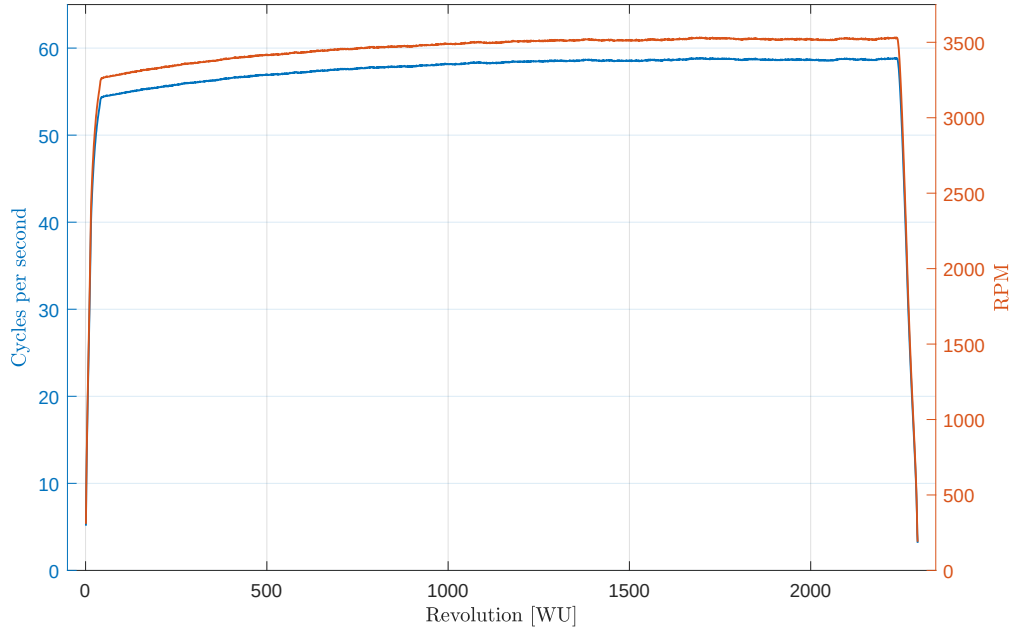


Figure 3: Rotation speed estimation.

The maximum shaft rotation speed is approximately 58.9 cycles per second, or 3,532 RPM. This maximum rotation speed occurs in revolution 1,672.

Spectrogram of Recording at the 0° Recording Angle

Figure 4 shows the spectrogram for the pressure recording for a measurement angle of 0°.

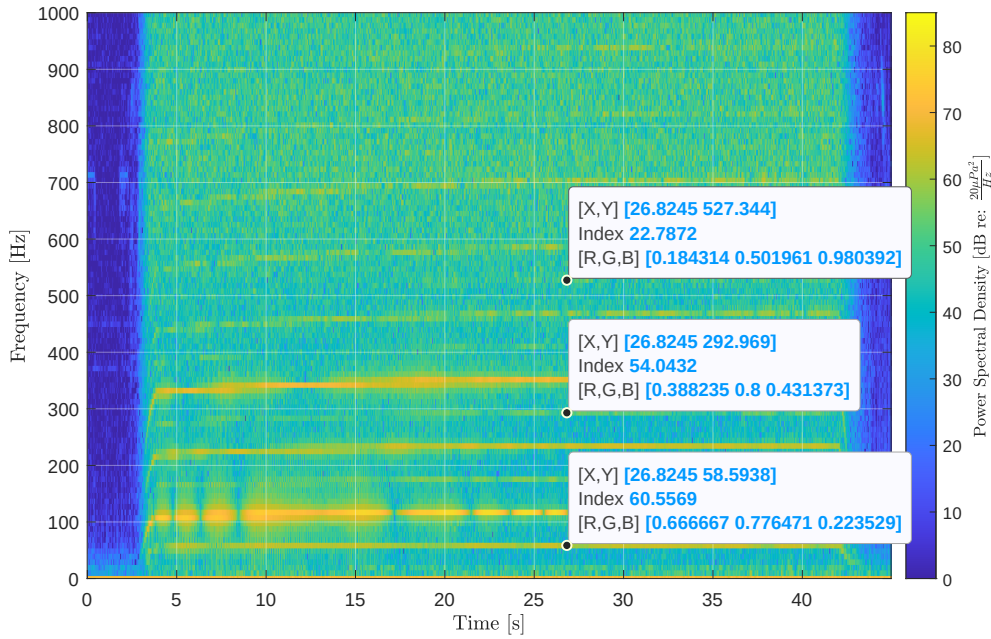


Figure 4: Spectrogram for the pressure recording at 0°.

The bottom trace in the spectrogram, which corresponds to the shaft rotation, is about 58.6 cycles per second.

1 Appendix - Matlab Code for Problem 1

```
1
2
3
4 %% Synopsis
5
6 % Problem 5 — Order Tracking of Mobius Propeller Data
7
8 % See Lecture 26
9 % "D:\15 Downloads\00 GitHub\ACS_547\35 Lectures\26 Monday, April 21, 2025\Lecture 26 — Shafting and
10 % bearings.pptx""
11
12
13 %% Note(s)
14
15 % Angle position is encoded here using a once-per-revolution pulse.
16
17
18
19 %% To Do
20
21 % Include labels on plots for each harmonic order (of blade rotation rate).
22
23
24
25 %% Environment
26
27 close all; clear; clc;
28 % restoredefaultpath;
29
30 addpath( genpath( './00 Support' ), '-begin' );
31
32 % set( groot, 'DefaultFigurePosition', [ 230 730 750 500 ] ); % x, y, width, height
33
34 set( 0, 'DefaultFigurePaperPositionMode', 'manual' );
35 set( 0, 'DefaultFigureWindowStyle', 'normal' );
36 set( 0, 'DefaultLineLineWidth', 1.0 );
37 set( 0, 'DefaultTextInterpreter', 'Latex' );
38
39 format ShortG;
40
41 color_map = slanCM( 'ColorBlind', 4 );
42
43 pause( 1 );
44
45
46
47 %% Load Data
48
49 load( 'mobius_prop_data.mat' ); % Variable(s): fs; mic_angles_degrees; p; trigger
50
51 % fs — Sample rate of 80 kHz (scalar).
52 % mic_angles_degrees — Microphone angles relative to horizontal plane (11-by-1; 23, 18, 6, 0, -6, -12,
53 % -18, -23, -28, and -32).
54 % p — Recorded pressures for the corresponding microphone angles (3,600,000-by-11).
55 % trigger — Shaft rotation synchronization signal (3,600,000-by-11).
56
57
58 %% RMS dB SPL Per Pressure Recording
59
60 % p_rms = 20*log10( rms( p ) / 20e-6 );
61 %
62 % figure( 'Name', 'RMS dB SPL Per Pressure Recording' ); ...
63 % stem( mic_angles_degrees, fliplr( p_rms ) ); grid on;
64 % xlabel( 'Microphone Angle [ $^{\circ}$ ]' );
65 % xticks( fliplr( mic_angles_degrees ) );
66 % ylabel( 'Sound Pressure [dB SPL]' );
67
68
69
70 %% Pressure Recordings
71
72 % figure( 'Name', 'Pressure Recordings' ); ...
73 % plot( p( :, 1 ) ); grid on;
74 % xlabel( 'Time [WU]' ); ylabel( 'Sound Pressure [Pa]' );
75
76
77
78 %% Visualize Experimental Data
79
80 net_time = size( p, 1 ) / fs; % 45 seconds
81
82 time_indices = ( 0:1:( numel( trigger ) - 1 ) ) ./ fs;
83 time_indices = time_indices(:);
84
85
86 % figure( 'Name', 'Trigger signal and Data for Horizontal Plane' ); ...
87 %
```

```

88 %     h1 = subplot( 2, 1, 1 ); ...
89 %         plot( time_indices, trigger, 'Color', color_map( 1, : ) ); grid on;
90 %         xlabel( 'Time [s]' ); ylabel( 'Voltage [V]' ); title( 'Shaft Trigger Signal' );
91 %         ylim( [ -0.5 4 ] );
92 %     h2 = subplot( 2, 1, 2 ); ...
93 %         plot( time_indices, p( :, 5 ), 'Color', color_map( 2, : ) ); grid on;
94 %         xlabel( 'Time [s]' ); ylabel( 'Pressure [Pa]' ); title( 'Recorded Pressure at 0$\circ$
Elevation' );
95 %         ylim( [ -5 5.5 ] );
96 %
97 %     linkaxes( [ h1 h2 ], 'x' );
98 %     xlim( [ -5 50 ] );
99
100
101
102 %% Spectrogram
103
104 signal = p(:, 5);
105
106 frame_length = 8192;
107 frame_hop = floor( 0.20 * frame_length );
108
109 a_spectrogram = spectrogram_September_26_2023( signal, frame_length, frame_hop, fs, 0 );
110
111 figure( 'Name', 'Spectrogram of Pressure Signal' ); ...
112
113 imagesc( a_spectrogram.time_indices, a_spectrogram.Sxx.frequencies, 20*log10( sqrt(a_spectrogram.
Sxx.spectrum)./20e-6 ), [ 0 85 ] );
114 labelColorbar( 'Power Spectral Density [dB re: $\frac{20}{\mu}$ Pa$^2$]{Hz}$' ); grid on;
115 colormap parula; % Option: Turbo
116 set( gca, 'GridColor', 'w', 'GridAlpha', 0.4 );
117 xlabel( 'Time [s]' ); ylabel( 'Frequency [Hz]' );
118 set( gca, 'ydir', 'normal' );
119 ylim( [ 0 1e3 ] );
120
121
122
123 %% Locate the Leading Edges of the Trigger Data
124
125 % Smooth data to reduce noise; supports finding the leading edges.
126 smoothed_trigger_data = [ 0; diff( smooth( trigger, 'moving', 5 ) ) ];
127
128 % Threshold smoothed data to located leading edges.
129 threshold = 0.5;
130 edge_indices = find( threshold < smoothed_trigger_data );
131 leading_edges = edge_indices( 1:2:end );
132
133 temp = diff( leading_edges );
134 [ indices_to_remove, indices ] = find( temp == 2 );
135 clear temp;
136
137 temp = leading_edges;
138 temp( indices_to_remove ) = [ ];
139
140 leading_edges = temp;
141 clear temp;
142
143 time_indices_leading_edge = leading_edges ./ fs;
144
145
146 figure( ); ...
147
148 plot( trigger( 1:1:end ), 'Color', 'k' ); hold on;
149 %
150 for index = 1:1:numel( leading_edges )
151     line( [ leading_edges( index ) leading_edges( index ) ], [ 0 4 ], 'Color', 'b' ); grid on;
152 end
153
154 axis( [ 1 numel( trigger ) -0.5 4.5 ] );
155
156
157
158 %% Segment Microphone Recording Channels
159
160 % Channel 1: 0 degrees
161 % Channel 2: 18 degrees
162 % Channel 3: 12 degrees
163 % Channel 4: 6 degrees
164 % Channel 5: 0 degrees
165 % Channel 6: -6 degrees
166 % Channel 7: -12 degrees
167 % Channel 8: -18 degrees
168 % Channel 9: -23 degrees
169 % Channel 10: -28 degrees
170 % Channel 11: -32 degrees
171
172
173 start_indices = 1:1:( numel( leading_edges ) - 1 ); end_indices = 2:1:numel( leading_edges );
174
175 frame_set_indices = [ leading_edges( start_indices(:) ) leading_edges( end_indices(:) ) ];
176 number_of_revolutions = size( frame_set_indices, 1 );
177

```

```

178
179 for channel_index = 1:1:size( p, 2 )
180
181     for frame_index = 1:1:size( frame_set_indices, 1 )
182
183         channel_segments{ frame_index, channel_index } = p( frame_set_indices( frame_index, 1 ):1:
184             frame_set_indices( frame_index, 2 ), channel_index );
185
186         slope_theta = (2*pi) ./ size( channel_segments{ frame_index, channel_index }, 1 );
187         theta{ frame_index, channel_index } = slope_theta .* ( 0:1:size( channel_segments{
188             frame_index, channel_index }, 1 ) - 1 );
189
190     end
191 end
192
193
194 %% Compute Instantaneous RPM
195
196 for channel_index = 1:1:size( p, 2 )
197     for frame_index = 1:1:size( frame_set_indices, 1 )
198         rpm_estimate{ frame_index, channel_index } = 60 * ( 1 / ( numel( channel_segments{ frame_index,
199             channel_index } ) ./ fs ) );
200     end
201 end
202
203 % figure( ); ...
204 %
205 %     yyaxis left; ...
206 %     plot( [ rpm_estimate{ :, 1 } ] ./ 60 ); grid on;
207 %     ylabel( 'Cycles per second' );
208 %     ylim( [ 0 65 ] );
209 %
210 %     yyaxis right; ...
211 %     plot( [ rpm_estimate{ :, 1 } ] ); grid on;
212 %     ylabel( 'RPM' );
213 %
214 %     xlabel( 'Revolution [WU]' );
215 %
216 %     axis( [ -50 number_of_revolutions+50 0 3.75e3 ] );
217 %     shg;
218
219
220
221 %% Compute Fourier Coefficients
222
223 An = nan( size( frame_set_indices, 1 ), 11 ); Bn = An;
224
225
226 TRACKING_ORDER = 1;
227
228 for channel_index = 1:1:size( p, 2 )
229
230     for frame_index = 1:1:size( frame_set_indices, 1 )
231
232         M = numel( channel_segments{ frame_index, channel_index } );
233         m = 0:1:( M - 1 );
234
235         An( frame_index, channel_index ) = ...
236             2/M * channel_segments{ frame_index, channel_index }.' * cos( TRACKING_ORDER .* theta{
237                 frame_index, channel_index } ).';
238
239         Bn( frame_index, channel_index ) = ...
240             2/M * channel_segments{ frame_index, channel_index }.' * sin( TRACKING_ORDER .* theta{
241                 frame_index, channel_index } ).';
242
243     end
244 end
245
246 figure( ); ...
247
248 h1 = subplot( 3, 1, 1 ); ...
249 plot( nan, nan ); hold on;
250 %
251 for channel_index = 1:1:size( p, 2 )
252     plot( An( :, channel_index ), 'Color', 'b' );
253 end
254 %
255 xlabel( 'Revolution [WU]' ); ylabel( 'An' );
256 %
257 grid on; axis( [ 1 number_of_revolutions -1 1 ] );
258
259 h2 = subplot( 3, 1, 2 ); ...
260 plot( nan, nan ); hold on;
261 %
262 for channel_index = 1:1:size( p, 2 )
263     plot( Bn( :, channel_index ), 'Color', 'r' ); grid on;
264 end

```

```

265 %
266 xlabel( 'Revolution [WU]' ); ylabel( 'Bn' );
267 %
268 grid on; axis( [ 1 number_of_revolutions -1 1 ] );
269
270 h3 = subplot( 3, 1, 3 ); ...
271 plot( nan, nan ); hold on;
272 %
273 for channel_index = 1:size( p, 2 )
274     plot( 20*log10( sqrt( An( :, channel_index ).^2 + Bn( :, channel_index ).^2 ) ./ 20e-6 ), '
        Color', 'k' );
275 end
276 %
277 xlabel( 'Revolution [WU]' ); ylabel( 'Sound Pressure dB re:20e-6 Pa' );
278 %
279 grid on; axis( [ 1 number_of_revolutions 50 110 ] );
280
281 shg;
282
283
284
285 %% Clean-up
286
287 % if ( ~isempty( findobj( 'Type', 'figure' ) ) )
288 %     monitors = get( 0, 'MonitorPositions' );
289 %     if ( size( monitors, 1 ) == 1 )
290 %         autoArrangeFigures( 3, 4, 1 );
291 %     elseif ( 1 < size( monitors, 1 ) )
292 %         autoArrangeFigures( 2, 2, 2 );
293 %     end
294 % end
295
296 fprintf( 1, '\n\n*** Processing Complete ***\n\n' );
297
298
299
300 %% Reference(s)
301
302 % https://www.mathworks.com/help/signal/ref/ordertrack.html#bvaw5ra-1-x
303
304 % https://www.mathworks.com/matlabcentral/answers/1439884-evaluate-rising-edge-sample-of-a-signal

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